Decision trees. A decision tree splits the data with a sequence of questions about single features. At each node, the algorithm chooses the split that most reduces impurity. Entropy measures the uncertainty in a set of labels, and information gain is the drop in entropy caused by a split. The Gini impurity is a similar measure that is faster to compute and is the default in many libraries. Trees are easy to interpret and can capture non-linear interactions, but a fully grown tree tends to overfit. Pruning limits the depth of the tree or removes branches that add little accuracy on validation data. Trees also handle mixed feature types and need little feature scaling.

Ensemble methods. An ensemble combines many models to get better predictions than any single model. Bagging trains each model on a bootstrap sample, which is a random sample drawn with replacement from the training set, and averages their predictions. Random forests extend bagging by also restricting each split to a random subset of features, which makes the individual trees less correlated. Boosting works differently: models are trained one after another, and each new model focuses on the examples the previous models got wrong. Gradient boosting fits each new tree to the negative gradient of the loss, the residual errors of the current ensemble. Gradient boosting machines are strong on tabular data, though they usually need more careful tuning than random forests.